Estimating the Causal Impact of Job Training Programs on Earnings

Evidence from the Lalonde National Supported Work Dataset

Jordi Visser ECON3916: Statistical and Machine Learning in Economics

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Introduction

The purpose of government job training programs is to improve participants' economic outcomes. However, it remains one of the biggest questions regarding labor economics. American policymakers allocate billions of dollars each year to develop the country's workforce, yet its effectiveness remains debatable. The challenge with these programs is the people enrolling. Individuals who enroll in job training are not randomly selected. They are specifically individuals with lower earnings, low interest in work, and fewer opportunities than the average person. This makes a comparison of earnings between participants and non-participants difficult, because it will result in a biased estimate of the program's effects.

This paper aims to address this by using the National Supported Work (NSW) Demonstration dataset, originally analyzed by Robert Lalonde in his 1986 study. The NSW

program was a job training program in which the government provided people with work experience, counseling, and support for a 9 to 18-month duration.

The central question of this paper is: Does participation in the National Supported Work job training program causally increase post-program earnings? The answer to this question will explain how these programs are designed and funded, particularly for people at a disadvantage with limited labor access.

Data Description

Source & Sample

The dataset used for this analysis is the Lalonde (1986) National Supported Work Demonstration dataset, sourced on Kaggle. The sample size of the dataset is 614 observations and 10 variables, which covers participants in the NSW job training program (the treatment group), compared against non-participants (the control group).

Variable Map

The dependent variable in this dataset is ‘re78’, which is real earnings measured in dollars, in 1978. This number represents the post-program labor market outcome, how much each participant earned in the year after the training program had ended. The independent variable is treat, indicating with ‘1’ if the individual participated in the NSW program, and a ‘0’ if the individual did not participate. The control variables consist of age (indicative of labor market experience), educ (years of education, indicative of human capital), re74 and re75 (pre-treatment earnings in 1974 and 1975), and demographic indicators such as black, hispan, married, and nodegree.

Data Audit

After auditing the dataset, it was confirmed to contain no missing values across all 11 columns, verified using a missing value diagnostic heatmap, and by proving there is no missing data, it helps eliminate a common potential source of bias in studies.

Using a histogram, the distribution of the dependent variable ‘re78’ was examined thoroughly, which revealed a clear right skew of wage distributions. A very large group of observations sits near zero, which reflects the high rate of unemployment, or very low earnings among program participants and the comparison group alike.

Table 1: Summary Statistics by Treatment Status

The table below displays mean values for key variables by treatment status, showing pre-treatment balance and post-treatment differences.

Table 1: Control Group

	re78	re74	re75	age	educ
count	429.000000	429.000000	429.000000	429.000000	429.000000
mean	6984.169742	5619.236506	2466.484443	28.030303	10.235431
std	7294.161791	6788.750796	3291.996183	10.786653	2.855238
min	0.000000	0.000000	0.000000	16.000000	0.000000
25%	220.181300	0.000000	0.000000	19.000000	9.000000
50%	4975.505000	2547.047000	1086.726000	25.000000	11.000000
75%	11688.820000	9277.128000	3881.419000	35.000000	12.000000
max	25564.670000	25862.320000	18347.230000	55.000000	18.000000

	black	hispan	married	nodegree
count	429.000000	429.000000	429.000000	429.000000
mean	0.202797	0.142191	0.512821	0.596737
std	0.402552	0.349654	0.500419	0.491126
min	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000
50%	0.000000	0.000000	1.000000	1.000000
75%	0.000000	0.000000	1.000000	1.000000
max	1.000000	1.000000	1.000000	1.000000

Table 2: Treated Group

	re78	re74	re75	age	educ
count	185.000000	185.000000	185.000000	185.000000	185.000000
mean	6349.143530	2095.573689	1532.055314	25.816216	10.345946
std	7867.402218	4886.620353	3219.250870	7.155019	2.010650
min	0.000000	0.000000	0.000000	17.000000	4.000000
25%	485.229800	0.000000	0.000000	20.000000	9.000000
50%	4232.309000	0.000000	0.000000	25.000000	11.000000
75%	9642.999000	1291.468000	1817.284000	29.000000	12.000000
max	60307.930000	35040.070000	25142.240000	48.000000	16.000000

	black	hispan	married	nodegree
count	185.000000	185.000000	185.000000	185.000000
mean	0.843243	0.059459	0.189189	0.708108
std	0.364558	0.237124	0.392722	0.455867
min	0.000000	0.000000	0.000000	0.000000
25%	1.000000	0.000000	0.000000	0.000000
50%	1.000000	0.000000	0.000000	1.000000
75%	1.000000	0.000000	0.000000	1.000000
max	1.000000	1.000000	1.000000	1.000000

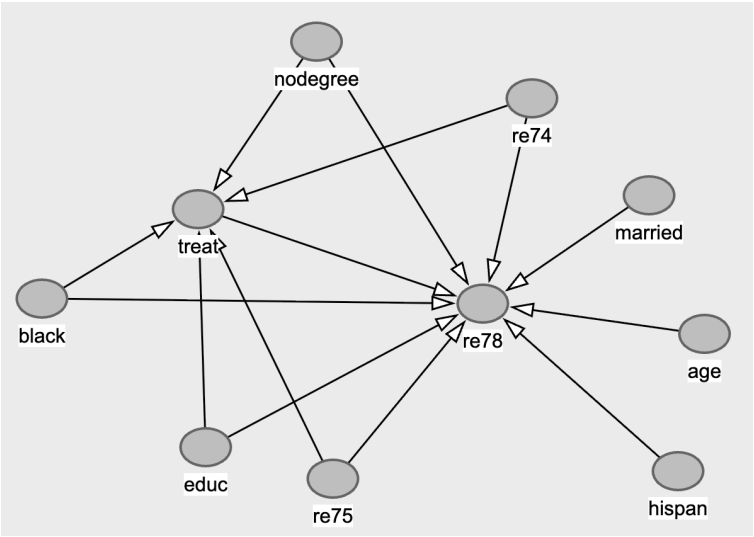
Observations worth noting: Treated individuals had much lower pre-treatment earnings in both re74 and re75, which confirms the selection problem mentioned in the introduction, that participants were economically disadvantaged compared to the control group before the program began. Treated individuals are also younger on average and less likely to be married. Most importantly, despite having lower pre-treatment earnings, post-treatment earnings, re78, are relatively similar across groups, suggesting a possible positive treatment effect once pre-treatment differences are controlled for.

Methodology

Identification Strategy

Ordinary Least Squares (OLS) regression was the primary identification strategy used for this paper, with a large set of pre-treatment control variables. After controlling for pre-treatment earnings, demographics, and education, we can safely assume that any other differences between the two groups are due to random chance rather than systematic selection. This assumption is reasonable because the NSW program selected participants based on observable characteristics that are measurable.

DAG



Regression Equation

$re78 = \beta_0 + \beta_1 treat + \beta_2 age + \beta_3 educ + \beta_4 re74 + \beta_5 re75 + \beta_6 black + \beta_7 hispan + \beta_8 married + \beta_9 nodegree + \varepsilon$

For this equation, re78 is the outcome variable, treat is the key independent variable, whose coefficient β_1 shows the causal effect of job training on earnings, and the rest of the terms are control variables used to isolate the true effect of job training. The final term 'ε' captures all unobserved factors.

Functional Form Justification

A Level-Level specification was chosen over a log transformation for two main reasons. One, re78 contains a large number of zeroes, which are participants with no 1978 earnings, making log(re78) undefined. Two, in Lalonde's original analysis, this same specification was used, allowing direct comparison with the benchmark findings.

Variable Justification

Pre-treatment earnings are included to control for prior participation in the labor market, which is the primary reason for selection into the program. Education and age are both indicative

of human capital and labor market experience. Race (black, hispan), marriage status (married), and degree status (nodegree) show demographics that independently affect labor market outcomes.

Robust Standard Errors

All models are estimated using HC1 Heteroskedasticity-Robust Standard Errors to ensure fair inference, because higher earners will have greater variance in their earnings than lower earners, which violates the standard OLS assumption of constant error variance.

Results

Variable	Baseline Model	Multivariate Model	Robust Model
treat	-635.03 (676.75)	1548.24**(740.58)	1100.30 (695.14)
age	-	12.98 (31.70)	10.37 (30.47)
educ	-	403.94** (160.40)	350.21** (151.21)
re74	-	0.296*** (0.071)	0.280*** (0.068)
re75	-	0.232 (0.119)	0.226 (0.117)
black	-	-1240.64 (695.77)	-1454.06** (686.54)
hispan	-	498.90 (952.86)	452.50 (948.39)
married	-	406.62 (658.55)	472.97 (651.85)
nodegree	-	259.82 (891.79)	138.85 (821.07)
N	614	614	611
R-squared	0.002	0.148	0.157

**p<0.10, **p<0.05, ***p<0.01. HC1 robust standard errors in parentheses.*

Treatment Effect: Participation in the NSW job training program is associated with a $1,548 increase in 1978 earnings, ceteris paribus (p=0.0037)

Education: An additional year of education is associated with a $404 increase in 1978 earnings, ceteris paribus (p=0.012).

Pre-Treatment Earnings (re74): Each additional dollar earned in 1974 is associated with $0.30 more in 1978 earnings, ceteris paribus (p=0.000).

Race: Being Black is associated with $1,241 lower earnings in 1978, ceteris paribus, significant at the 10% level (p=0.075)

Robustness Checks

The primary robustness check in the analysis involves re-estimating the baseline model after removing any outliers identified using the Turkey Fence method. Observations where re78 exceeded Q3 + 1.5 * IQR were removed, reducing the sample size from 614 to 611.

After the outliers were removed, the treatment effect coefficient fell from $1,548 to $1,100. Despite the dollar amount decreasing, the direction of the effect is still clear: job training is positively associated with earnings. Along with that, the coefficients on education and re74 remain significant across both tests, confirming the two are not driven by outliers.

While there is a reduction in treatment effect once outliers are removed, due to a small number of high earners inflating the baseline estimate, the finding that treatment remains positive in both cases indicates a positive correlation.

Conclusion

This paper estimated the causal impact of participation in the National Supported Work job training program on 1978 earnings using OLS regression with multiple pre-treatment controls. The baseline estimate suggests that people who participated in the program increased earnings by approximately $1,548, which is found to be statistically significant at p=0.0037. This finding can allow for the cautious conclusion that job training programs can improve the economic outcomes of disadvantaged workers.

However, there are a few limitations to this. First, the model assumes that all important differences between the treated and control groups are shown by the variables. There may be unobserved factors affecting earnings, like motivation or mental state, that could overstate the treatment effect. Second, an R-squared of 0.148 means that the model only explains ~15% of the variation in earnings, meaning many factors are not shown in the dataset that could affect labor market outcomes. Finally, the NSW program was in the 1970s, and the labor market then was significantly different from the one now.

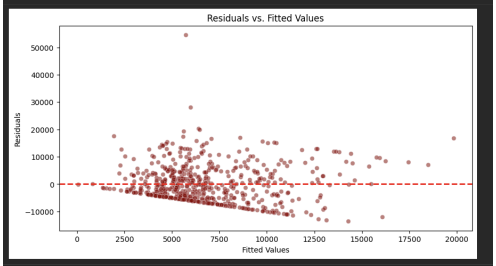
Code Appendix

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.formula.api as smf

df = pd.read_csv('lalonde.csv')
```

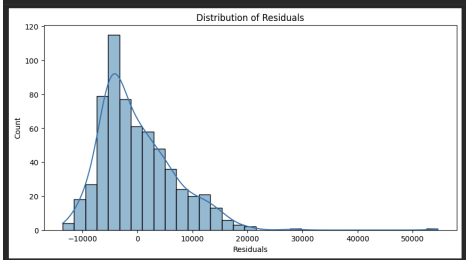
```python
# Step 4.1
formula_final = 're78 ~ treat + age + educ + re74 + re75 + black + hispan + married + nodegree'

model_final = smf.ols(formula = formula_final, data=df).fit(cov_type='HC1')
print(model_final.summary())
formula_baseline = 're78 ~ treat'
model_baseline = smf.ols(formula=formula_baseline, data=df).fit(cov_type='HC1')
print(model_baseline.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   re78   R-squared:                       0.148
Model:                            OLS   Adj. R-squared:                  0.135
Method:                 Least Squares   F-statistic:                     9.957
Date:                Thu, 26 Mar 2026   Prob (F-statistic):           2.55e-14
Time:                        00:21:02   Log-Likelihood:                -6297.8
No. Observations:                 614   AIC:                         1.262e+04
Df Residuals:                     604   BIC:                         1.266e+04
Df Model:                           9
Covariance Type:                  HC1
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     66.5145   2530.018      0.026      0.979   -4892.231    5025.260
treat       1548.2438    740.576      2.091      0.037      96.741    2999.746
age           12.8776     31.704      0.409      0.682     -49.161      75.116
educ         403.9412    160.395      2.518      0.012      89.574     718.309
re74           0.2964      0.071      4.159      0.000       0.157       0.436
re75           0.2315      0.119      1.938      0.053      -0.003       0.466
black      -1248.6441    695.766     -1.783      0.075   -2604.321     123.032
hispan       498.8969    952.856      0.524      0.601   -1368.667    2366.461
married      406.6200    658.549      0.617      0.537    -884.112    1697.354
nodegree     259.8174    891.786      0.291      0.771   -1488.050    2007.685
==============================================================================
Omnibus:                      216.123   Durbin-Watson:                   2.009
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1182.664
Skew:                           1.467   Prob(JB):                    1.54e-257
Kurtosis:                       9.134   Cond. No.                     7.64e+04
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
[2] The condition number is large, 7.64e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
                            OLS Regression Results
==============================================================================
Dep. Variable:                   re78   R-squared:                       0.002
Model:                            OLS   Adj. R-squared:                 -0.000
Method:                 Least Squares   F-statistic:                    0.8885
Date:                Thu, 26 Mar 2026   Prob (F-statistic):              0.348
Time:                        00:21:02   Log-Likelihood:                -6346.4
No. Observations:                 614   AIC:                         1.270e+04
Df Residuals:                     612   BIC:                         1.271e+04
Df Model:                           1
Covariance Type:                  HC1
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept   6984.1697    352.329     19.823      0.000    6293.617    7674.722
treat       -635.0262    676.740     -0.938      0.348   -1961.420     691.375
==============================================================================
Omnibus:                      209.619   Durbin-Watson:                   1.809
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              775.567
Skew:                           1.567   Prob(JB):                    3.87e-169
Kurtosis:                       7.527   Cond. No.                         2.42
==============================================================================
```

```python
# Step 4.2
# Diagnostic 1
fitted = model_final.fittedvalues
residuals = model_final.resid

plt.figure(figsize=(10,5))
sns.scatterplot(x=fitted, y=residuals, alpha=0.5, color='darkred')
plt.axhline(0, color='red', linestyle='--', linewidth=2)
plt.title('Residuals vs. Fitted Values')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.show()
```




```python
# Diagnostic 2
plt.figure(figsize=(10,5))
sns.histplot(residuals, kde=True)
plt.title('Distribution of Residuals')
plt.xlabel('Residuals')
plt.show()
```



```python
# Diagnostic 3
print("Table 1: Control Group")
print(df[df['treat'] == 0][['re78', 're74', 're75', 'age', 'educ', 'black', 'hispan', 'married', 'nodegree']].describe())
print("Table 2: Treated Group")
print(df[df['treat'] == 1][['re78', 're74', 're75', 'age', 'educ', 'black', 'hispan', 'married', 'nodegree']].describe())
```

```
Table 1: Control Group
                re78         re74         re75         age       educ  \
count    429.000000   429.000000   429.000000  429.000000  429.000000
mean    6984.169742  5619.236506  2466.484443   28.030303   10.235431
std     7294.161791  6788.750796  3291.996183   10.786653    2.855238
min        0.000000     0.000000     0.000000   16.000000    0.000000
25%      220.181300     0.000000     0.000000   19.000000    9.000000
50%     4975.505000  2547.047000  1086.726000   25.000000   11.000000
75%    11688.820000  9277.128000  3881.419000   35.000000   12.000000
max    25564.670000 25862.320000 18347.230000   55.000000   18.000000

             black      hispan     married    nodegree
count  429.000000  429.000000  429.000000  429.000000
mean     0.202797    0.142191    0.512821    0.596737
std      0.402552    0.349654    0.500419    0.491126
min      0.000000    0.000000    0.000000    0.000000
25%      0.000000    0.000000    0.000000    0.000000
50%      0.000000    0.000000    1.000000    1.000000
75%      0.000000    0.000000    1.000000    1.000000
max      1.000000    1.000000    1.000000    1.000000
Table 2: Treated Group
                re78         re74         re75         age       educ  \
count    185.000000   185.000000   185.000000  185.000000  185.000000
mean    6349.143530  2095.573689  1532.055314   25.816216   10.345946
std     7867.402218  4886.620353  3219.250870    7.155019    2.010650
min        0.000000     0.000000     0.000000   17.000000    4.000000
25%      485.229800     0.000000     0.000000   20.000000    9.000000
50%     4232.309000     0.000000     0.000000   25.000000   11.000000
75%     9642.999000  1291.468000  1817.284000   29.000000   12.000000
max    60307.930000 35040.070000 25142.240000   48.000000   16.000000

             black      hispan     married    nodegree
count  185.000000  185.000000  185.000000  185.000000
mean     0.843243    0.059459    0.189189    0.705405
std      0.364558    0.237124    0.392722    0.455867
min      0.000000    0.000000    0.000000    0.000000
25%      1.000000    0.000000    0.000000    0.000000
50%      1.000000    0.000000    0.000000    1.000000
75%      1.000000    0.000000    0.000000    1.000000
max      1.000000    1.000000    1.000000    1.000000
```

```python
# Diagnostic 4

Q1 = df['re78'].quantile(0.25)
Q3 = df['re78'].quantile(0.75)
IQR = Q3 - Q1
outliers = df[(df['re78'] < (Q1-1.5*IQR)) | (df['re78'] > (df['re78'] > (Q3 + 1.5 * IQR)))]
print(f"Outliers removed: {len(outliers)}")

df_clean = df[df['re78'] <= (Q3 + 1.5 * IQR)]
model_robust = smf.ols(formula=formula_final, data=df_clean).fit(cov_type='HC1')

print(f"Original treat coef: {model_final.params['treat']:.2f}")
print(f"Robust treat coef: {model_robust.params['treat']:.2f}")
print(model_robust.summary())
```

```
Outliers removed: 471
Original treat coef: 1548.24
Robust treat coef: 1100.30
                            OLS Regression Results
==============================================================================
Dep. Variable:                   re78   R-squared:                       0.157
Model:                            OLS   Adj. R-squared:                  0.144
Method:                 Least Squares   F-statistic:                     10.42
Date:                Thu, 26 Mar 2026   Prob (F-statistic):           4.79e-15
Time:                        00:21:03   Log-Likelihood:                -6221.9
No. Observations:                 611   AIC:                         1.246e+04
Df Residuals:                     601   BIC:                         1.251e+04
Df Model:                           9
Covariance Type:                  HC1
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept    885.6155   2367.254      0.374      0.708   -3754.116    5525.348
treat       1100.3049    695.142      1.583      0.113    -262.148    2462.758
age           10.3716     30.472      0.340      0.734     -49.353      70.096
educ         350.2080    151.209      2.316      0.021      53.843     646.573
re74           0.2795      0.068      4.113      0.000       0.146       0.413
re75           0.2262      0.117      1.932      0.053      -0.003       0.456
black      -1454.0622    686.541     -2.118      0.034   -2799.658    -108.466
hispan       452.5042    948.393      0.477      0.633   -1406.312    2311.321
married      472.9666    651.848      0.726      0.468    -804.633    1750.566
nodegree     138.8536    821.066      0.169      0.866   -1470.407    1748.114
==============================================================================
Omnibus:                       47.437   Durbin-Watson:                   1.975
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               57.222
Skew:                           0.749   Prob(JB):                     3.75e-13
Kurtosis:                       3.061   Cond. No.                     7.52e+04
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
[2] The condition number is large, 7.52e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

```python
print("\n Robustness Check: Model without Outliers")
print(f"Original treat coef: {model_final.params['treat']:.2f}")
print(f"Robust treat coef: {model_robust.params['treat']:.2f}")
print("\n Full Robust Model Summary")
print(model_robust.summary())
```

```
 Robustness Check: Model without Outliers
Original treat coef: 1548.24
Robust treat coef: 1100.30

 Full Robust Model Summary
                            OLS Regression Results
==============================================================================
Dep. Variable:                   re78   R-squared:                       0.157
Model:                            OLS   Adj. R-squared:                  0.144
Method:                 Least Squares   F-statistic:                     10.42
Date:                Thu, 26 Mar 2026   Prob (F-statistic):           4.79e-15
Time:                        00:21:03   Log-Likelihood:                -6221.9
No. Observations:                 611   AIC:                         1.246e+04
Df Residuals:                     601   BIC:                         1.251e+04
Df Model:                           9
Covariance Type:                  HC1
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept    885.6155   2367.254      0.374      0.708   -3754.116    5525.348
treat       1100.3049    695.142      1.583      0.113    -262.148    2462.758
age           10.3716     30.472      0.340      0.734     -49.353      70.096
educ         350.2080    151.209      2.316      0.021      53.843     646.573
re74           0.2795      0.068      4.113      0.000       0.146       0.413
re75           0.2262      0.117      1.932      0.053      -0.003       0.456
black      -1454.0622    686.541     -2.118      0.034   -2799.658    -108.466
hispan       452.5042    948.393      0.477      0.633   -1406.312    2311.321
married      472.9666    651.848      0.726      0.468    -804.633    1750.566
nodegree     138.8536    821.066      0.169      0.866   -1470.407    1748.114
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Omnibus:                       47.437   Durbin-Watson:                   1.975
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```